Problem Set 3

Data Visualisation for Social Scientists

Due: February 18, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Wednesday February 18, 2026. No late assignments will be accepted.

Canadian Election Study

The data for this problem set come from the Canadian Election Study (CES) in 2015. The main purpose of the study is to give a comprehensive picture of the Canadian election: why people vote as they do, what changes during campaigns and across elections, and how Canadian voting compares with that in other democracies.

Data Manipulation

1. Load the CES .csv file from GitHub into your global environment. Filter respondents to only include "high quality" participants:

```r
library(tidyverse)
ces_data <- read_csv("C:/Users/molly/OneDrive/Documents/GitHub/DataViz_2026/problemsets/PS03/my_answers/CES2015.csv")
ces_data <- ces_data |>
  filter(discard == "Good quality")
names(ces_data)
```

I loaded the csv file and filtered the dataset to include only high quality participants

```
ces2015 <- ces2015 |> filter(discard == "Good quality")
```

2. Filter the dataset to those participants that answered the question about voting for the past election using `p_voted`. Consider respondents who gave a "Yes" answer as having voted, while “No” as not having voted. Treat “Don’t know” and “Refused” as missing.

```r
ces_data <- ces_data |>
  mutate(
    voted_flag = case_when(
      p_voted == "Yes" ~ "Voted",
      p_voted == "No" ~ "Did not vote",
      p_voted %in% c("Dont know", "Refused") ~ NA_character_,
      TRUE ~ NA_character_
    )
  )
table(ces_data$voted_flag)
ces_data <- ces_data |>
  mutate(
    age_num = 2015 - as.numeric(age)
  )
```

I filtered the dataset to include only respondents who answered the question about voting in the past election. I coded those who said “Yes” as having voted, “No” as not having voted, and treated “Don’t know” and “Refused” as missing values so they wouldn’t interfere with my turnout analysis.

```
Did not vote        Voted
299          4415
```

3. Create an age variable and group into categories (e.g., <30, 30-44, 45-64, 65+). Year of birth is in age (four-digit year).

```r
ces_data <- ces_data |>
  mutate(
    age_group = cut(
      age_num,
      breaks = c(-Inf, 29, 44, 64, Inf),
      labels = c("<30", "30-44", "45-64", "65+")
    )
  )
table(ces_data$age_group, useNA = "ifany")
```

```
  <30 30-44 45-64   65+  <NA>
1414  2053  3812  2518   185
```

Created an age variable and grouped age into categories.

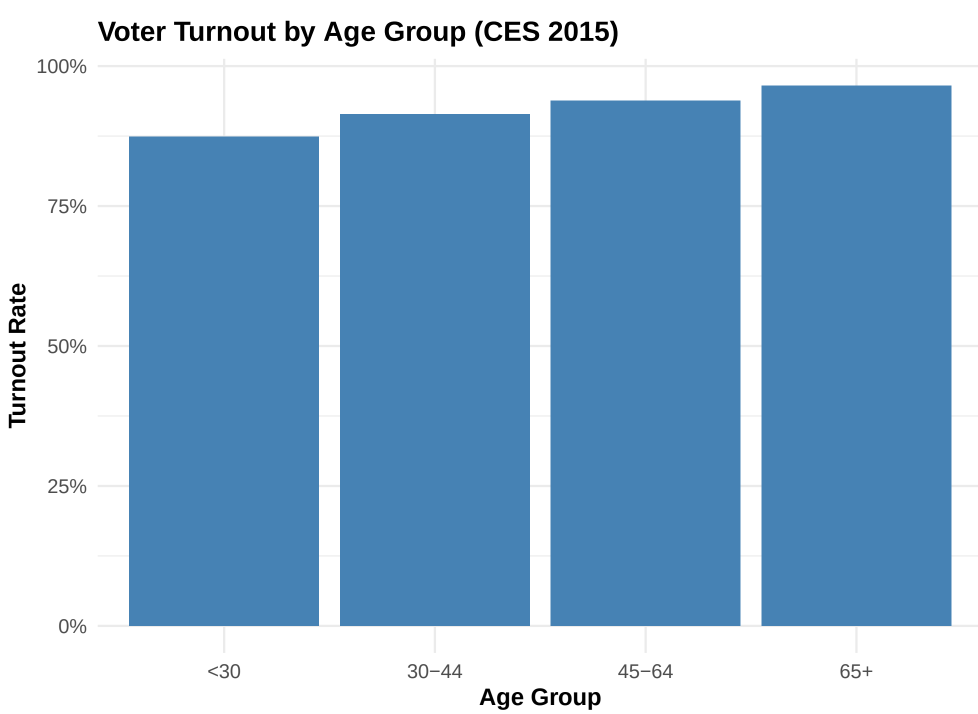
Data Visualization

1. Plot turnout rate by age group.

```r
library(ggplot2)
library(dplyr)
ces_turnout <- ces_data |>
  filter(!is.na(voted_flag) & !is.na(age_group))

turnout_by_age <- ces_turnout |>
  group_by(age_group) |>
  summarize(
    turnout_rate = mean(voted_flag == "Voted"),
    n = n()
  )
plot1 <- ggplot(turnout_by_age, aes(x = age_group, y = turnout_rate)) +
  geom_col(fill = "steelblue") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    x = "Age group",
    y = "Turnout rate",
    title = "Voter Turnout by Age Group (CES 2015)"
  ) +
  theme_minimal()
ggsave("graph1.pdf", plot = plot1, width = 7, height = 5)
```

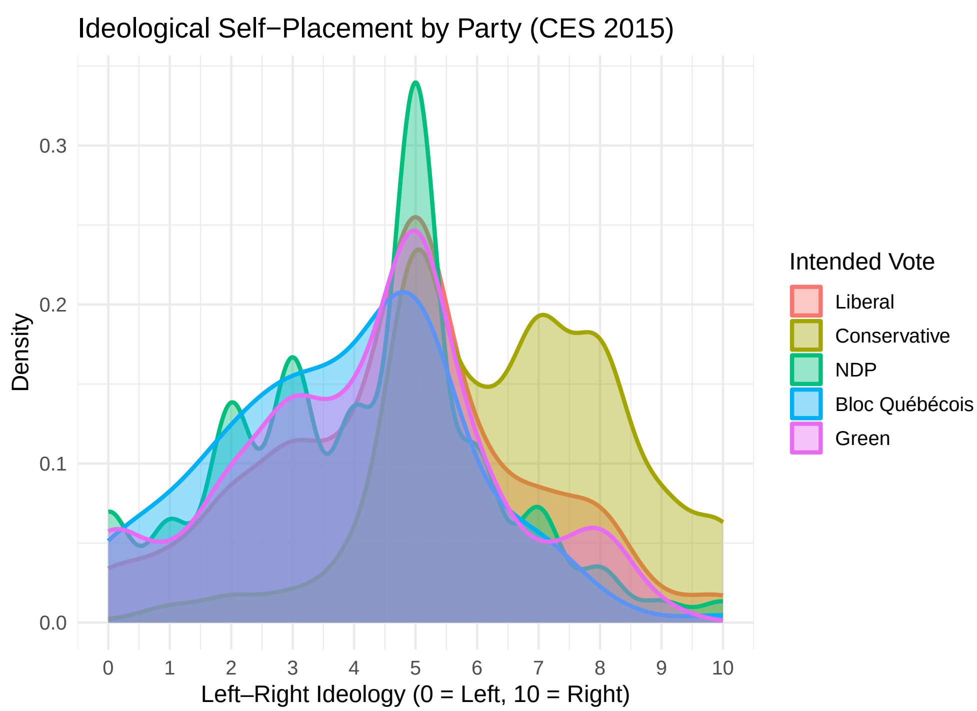
I summarized voter turnout by age group and plotted it as a bar chart.



2. Create a density plot of ideology by party, restricting your sample to respondents with non-missing left–right self-placement (0–10 scale) and those that intended to vote for a main party (e.g., Liberal, Conservative, NDP, Bloc in Quebec, and Green).

```
ces_ideo <- ces_ideo %>%
  mutate(vote_for = factor(vote_for,
                           levels = c("Liberal", "Conservative", "NDP", "Bloc Qu b cois", "Green")))
graph2 <- ggplot(ces_ideo, aes(x = p_selfplace, fill = vote_for, color = vote_for)) +
  geom_density(alpha = 0.4, size = 1) +
  scale_x_continuous(breaks = 0:10, labels = 0:10) +
  labs(
    x = " L e f t Right  Ideology (0 = Left, 10 = Right)",
    y = "Density",
    fill = "Intended Vote",
    color = "Intended Vote",
    title = "Ideological Self-Placement by Party (CES 2015)"
  ) +
  theme_minimal(base_size = 12)
ggsave("graph2.pdf", plot = graph2, width = 7, height = 5)
```

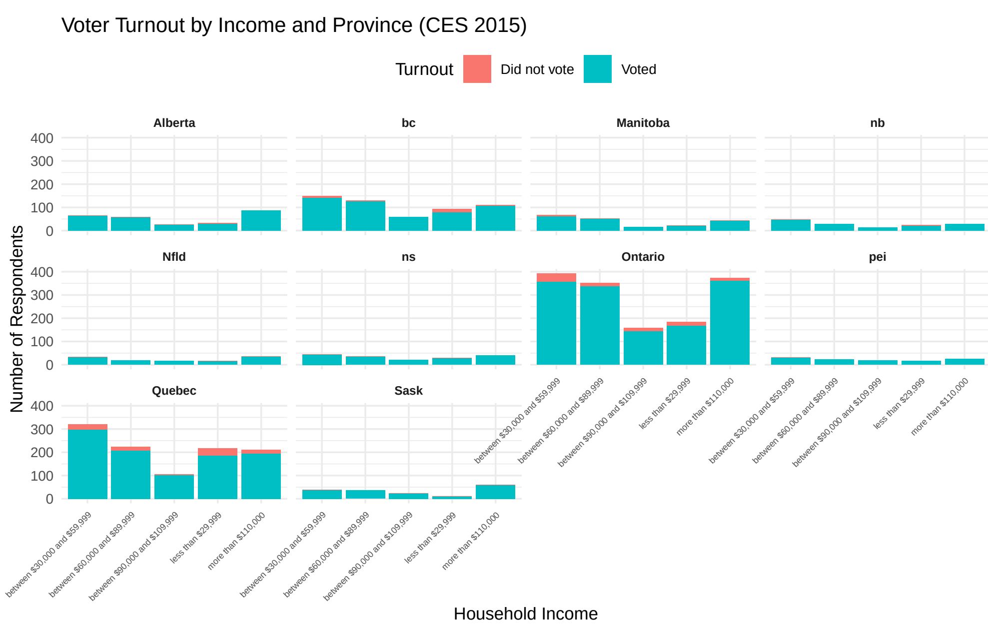
I converted respondents' ideology scores to numeric, kept only main parties, and plotted a density graph showing how left–right ideology is distributed by party.



3. Produce histogram counts of turnout by income (`income_full`), faceted by province.

```r
library(tidyverse)
ces_turnout_income <- ces_data |>
  filter(discard == "Good quality") |>
  filter(!is.na(voted_flag)) |>
  filter(!is.na(income_full)) |>
  filter(!income_full %in% c(".d", ".r"))
graph3 <- ggplot(ces_turnout_income, aes(x = income_full, fill = voted_flag)) +
  geom_bar(position = "stack") +
  facet_wrap(~ province, ncol = 4) +
  labs(
    title = "Voter Turnout by Income and Province (CES 2015)",
    x = "Household Income",
    y = "Number of Respondents",
    fill = "Turnout"
  ) +
  theme_minimal(base_size = 10) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, size = 5),  # extra small
    strip.text = element_text(face = "bold", size = 7),
    legend.position = "top"
  )
ggsave("graph3.pdf", plot = graph3, width = 8, height = 5)
```

I filtered the data to include only high-quality respondents with valid turnout and income information, excluding any invalid codes like ".d" and ".r". Then I plotted a stacked bar chart showing the number of respondents who voted or didn't vote across income categories, faceted by province.



4. Create your own reusable custom theme. Apply your theme to one of the previous plots and add:

   (a) An improved title summarizing the main substantive takeaway.

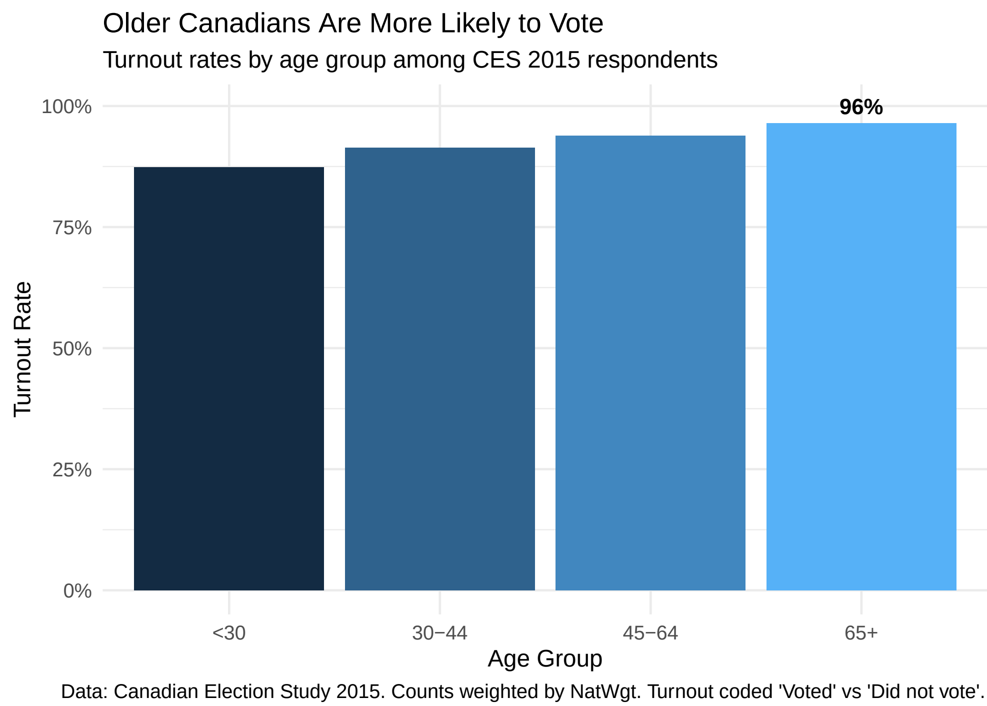
   (b) A more informative subtitle describing the sample and variables.

   (c) A caption noting data source, weighting, and key coding decisions.

   (d) At least one direct annotation using `ggrepel` that calls out a key pattern.

```r
graph4 <- ggplot(age_turnout_summary, aes(x = age_group, y = turnout_rate, fill = turnout_rate)) +
  geom_col(show.legend = FALSE) +
  scale_y_continuous(labels = scales::percent) +
  labs(
    title = "Older Canadians Are More Likely to Vote",
    subtitle = "Turnout rates by age group among CES 2015 respondents",
    x = "Age Group",
    y = "Turnout Rate",
    caption = "Data: Canadian Election Study 2015. Counts weighted by NatWgt. Turnout coded 'Voted' vs 'Did not vote'."
  ) +
  geom_text_repel(
    data = annotation_point,
    aes(
      x = age_group,
      y = turnout_rate,
      label = paste0(round(turnout_rate*100), "%")
    ),
    nudge_y = 0.03,
    size = 4,
    fontface = "bold",
    segment.color = "red"
  ) +
  theme_minimal(base_size = 12)
ggsave("graph4.pdf", plot = graph4, width = 7, height = 5)
```

I improved the original age-turnout bar chart by adding a more informative title and subtitle to summarize the main takeaway and describe the sample. I included a caption noting the data source, weighting, and coding decisions. I also highlighted the age group with the highest turnout using a ggrepel annotation and applied my custom theme to enhance the overall appearance.


Older Canadians Are More Likely to Vote
Turnout rates by age group among CES 2015 respondents
100%
75%
50%
25%
0%
Turnout Rate
96%
<30
30−44
45−64
65+
Age Group
Data: Canadian Election Study 2015. Counts weighted by NatWgt. Turnout coded 'Voted' vs 'Did not vote'.